# SemEval-2027 Task Proposal: Coherence-Aware Compositional Fine Grained Natural Language Inference (CoCo-NLI)

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## 1 Overview

**Summary of the task.** We propose **CoCo-NLI** (**Co**herence-aware **Co**mpositional **N**atural **L**anguage **I**nference), a proposal for a new shared task on reversal-aware NLI.

**Why this task is needed and who will participate.** Large benchmark averages are no longer sufficient to evaluate state-of-the-art LLMs: subtle reasoning failures often disappear inside broad aggregate metrics (Lopez-Gazpio, 2024). This is especially problematic for NLI (MacCartney, 2009), where systems can achieve strong headline scores while relying on shallow cues rather than robust semantic composition (Gururangan et al., 2018). Our motivation is that **specific evaluation metrics and fine-grained NLI is necessary** to expose the types of directional errors that vanish within hollow evaluation metrics (Apaolaza et al., 2025). *CoCo-NLI* directly targets one such logical inference challenge: the *Reversal Curse* (RC), i.e., the inability of LLMs –especially autoregressive decoders– to preserve inference when relation direction is reversed (Berglund et al., 2023).

*CoCo-NLI* reframes the RC in a genuine NLI setting rather than a narrow factual-retrieval setup, and therefore connects two active research lines that do not yet share a common benchmark: (i) semantic evaluation and NLI , and (ii) LLM, interpretability, and reasoning evaluation. The task should also attract multilingual NLP researchers, since the same inference phenomenon will be evaluated in English, Spanish, and Basque.

**Expected impact.** *CoCo-NLI* is **not a rerun**: it introduces a new semantic evaluation problem centered on *directional inferential coherence* (Apaolaza et al., 2026). Its impact is threefold. First, it provides the first shared task focused on whether models maintain inference consistency under reversal in fine-grained NLI. Second, it offers a reusable benchmark and scorer that remain meaningful after SemEval, enabling future work on evaluation, model construction, augmentation, debiasing, and error analysis. Third, by releasing multilingual data, baselines, and evaluation tools in a long-term repository, the task will lower the barrier to systematic work on reasoning failures beyond English and beyond closed industrial models.

**Briefing of the main task.** Given an *ordered* pair of short phrases derived from the PhrasIS dataset (Lopez-Gazpio et al., 2024), systems must predict one of three direction-sensitive labels: EQUIVALENCE, FORWARD-ENTAILMENT, or BACKWARD-ENTAILMENT. Unlike standard NLI evaluations, *CoCo-NLI* also tests whether a system remains *logically coherent* when the same phrase pair is presented in reversed order. The task will provide three language tracks: **English**, **Spanish**, and **Basque**. The official ranking will emphasize **soft causal coherence and hard causal coherence** under reversal (Apaolaza et al., 2026).

## 2 Data and Resources

**How the data will be produced.** The task reuses and extends two already validated resources. The starting point is **PhrasIS**, a benchmark of naturally occurring phrase pairs extracted from image captions and news headlines, annotated by experts for fine-grained inference and similarity (Lopez-Gazpio et al., 2024). From PhrasIS we select the subset of labels that can be reversed with high confidence: EQUIVALENCE, FORWARD-ENTAILMENT, and BACKWARD-ENTAILMENT. Following our pilot study at Apaolaza et al. (2026), each original phrase pair is paired with its reversed counterpart, and the reversed compatible gold label is assigned automatically through a deterministic label-reversal operator: FORWARD $\leftrightarrow$ BACKWARD, while EQUIVALENCE remains unchanged. This yields an evaluation set in which every item belongs to a minimal pair of logically linked instances.

For SemEval, we will release: (i) **Official task data**: the curated reversal-aware subset in EN/ES/EU. (ii) **Auxiliary data**: the broader English PhrasIS resource for participants who want extra supervision. (iii) **Starter kit**: format checker, official scorer, baseline systems, and CodaBench-ready submission templates.

The English track reuses the already curated pilot data. The Spanish and Basque tracks will be created by semiautomatic translation of the English task instances by deep machine translation and bilingual annotators familiar with fine-grained semantic distinctions.

**Copyright and license.** The newly created shared-task artifacts –task splits, reversal metadata, multilingual translations, scorer, baseline code, and documentation– will be released on Zenodo and GitHub under a research-friendly license (**CC BY 4.0**). Reused source material will follow the terms of the upstream resources. If any upstream restriction prevents redistribution of a full source component, we will distribute the task-ready derived phrase pairs together with provenance metadata and reconstruction scripts, so that the benchmark remains usable after the competition.

**How much data will be produced.** Published PhrasIS statistics contain 1,946 English source pairs with reversible labels (EQUIVALENCE, FORWARD, BACKWARD). After adding the reversed counterparts, the English official task is expected to contain roughly **4k ordered instances**. With Spanish and Basque translations, the final official release is expected to contain roughly **10k ordered instances after filtering** across languages.

**How data quality will be ensured and evaluated.** The English source data already comes from a benchmark with strong expert annotation quality. For PhrasIS, inter-annotator agreement on the relation labels was reported as 0.78/0.82 accuracy and 0.66/0.77 Cohen's $\kappa$ for Headlines/Images, with Pearson 0.79/0.80 for similarity scoring. For the SemEval release, all multilingual dev/test instances will be checked. We will also run consistency checks on the reversal operator and on label symmetry/asymmetry before release.

**Example.**

```
pair_id=1032, lang=en
premise=Everyone is hungry
hypothesis=Someone is hungry
label=Forward-Entailment,
reverse_label=Backward-Entailment
```

The reversed item swaps premise and hypothesis and must receive the compatible reverse label.

**Resources required.** Because the English pilot, metrics, and baselines already exist, the main remaining effort is multilingual translation, verification, packaging, and infrastructure. We estimate: (i) under 200 expert hours for ES/EU translation and adjudication, (ii) modest compute for retraining multilingual baselines, and (iii) standard shared-task infrastructure: website, mailing list, GitHub, Zenodo, and CodaBench.

**Ethics, privacy, and security.** The task uses phrase pairs from publicly available captions and news headlines. It contains no medical decision-making, no private-user profiling, and no task aimed at identifying specific people. We do not use personal conversations, sensitive records, or hidden metadata. The main ethical risk is not social harm from the task itself, but the possibility that benchmark systems overfit shallow cues; which is indeed the behavior the task is designed to reveal.

## 3 Pilot Task

Our proposal is backed by a strong pilot study by Apaolaza et al. (2026). In recent work, we already defined the English reversal-aware benchmark and evaluated a broad set of baseline systems across three architecture families: **encoder-only**, **decoder-only**, and **encoder–decoder**. The pilot includes modern open-weight baselines (e.g., DeBERTa-v3, ModernBERT, RoBERTa, OPT, Pythia, GPT-2, BART, Flan-T5), standard label-prediction scores, and the new coherence metrics. This makes *CoCo-NLI* low-risk from the organizational perspective and high-value from the scientific perspective.

**Lessons learned and how they shape the task design.** The pilot yielded four design lessons. First, standard label metrics alone hide important directional failures; therefore coherence must be an official evaluation target. Second, positive fine-grained distinctions are substantially more diagnostic than easier setups dominated by unrelated pairs; therefore the official task focuses on the reversible entailment/equivalence subset. Third, encoder models are currently stronger than decoder-only models, but they still confuse entailment with near-similarity and equivalence; therefore there is clear room for improvement. Fourth, hyperparameter sensitivity is high in this setting; to lower the barrier for participation, we will release strong baselines, training scripts, and an official scorer.

The SemEval version extends the pilot in two directions. First, it adds **Spanish** and **Basque** tracks, enabling multilingual and cross-lingual experimentation. Second, all submitted runs will be tagged by (a) **access regime**: open-weight vs. commercial/API-based, and (b) **architecture family**: encoder-only, decoder-only, encoder–decoder. These tags will not fragment the official leaderboard, but they will support the post-task analysis that the community currently lacks.

## 4 Evaluation

Each system must predict a label for every ordered phrase pair in the hidden test set. Evaluation will report three complementary measures: (i) **Weighted** $F_1$ for standard 3-way label prediction. (ii) **SoftCoh**: whether predictions for an item and its reversed counterpart are mutually compatible under the label-reversal operator, regardless of gold correctness. (iii) **HardCoh**: whether the predictions are both mutually compatible *and* correct with respect to the gold labels.

Formally, for an original item $x_i$ and its reversed counterpart $x_i^{rev}$, SoftCoh assigns 1 iff $f(x_i) = Rev(f(x_i^{rev}))$. HardCoh assigns 1 iff this coherence condition holds *and* both predictions match the corresponding gold labels. Track scores will be the average of these indicators over all paired instances.

**Official ranking.** The primary ranking metric will be **HardCoh**, because the core scientific question of the task is whether a model preserves *correct* inference under reversal. **Weighted** $F_1$ will be used as a tie-break and as an auxiliary metric. **SoftCoh** will be reported to separate directional self-consistency from correctness. Scores will be reported per language and as a multilingual macro-average across EN/ES/EU.

We will provide an official scorer, format validator, and reference implementation. The task will run on CodaBench, with one primary submission per language track and model type and a small number of additional contrastive submissions per team.

## 5 Task Organizers

**Inigo Lopez-Gazpio** (Lead Organizer), HiTZ Basque Center for Language Technology – Ixa NLP Group, University of the Basque Country UPV/EHU, `inigo.lopez@ehu.eus`. Expertise in semantic evaluation, STS, interpretable STS, fine-grained semantics, and benchmark design. Previous SemEval organizer/co-organizer experience includes **SemEval-2015 Task 2**, **SemEval-2016 Task 2**, and **SemEval-2017 Task 1**.

**Jon F. Apaolaza** (Co-Organizer), HiTZ Basque Center for Language Technology – Ixa NLP Group, University of the Basque Country UPV/EHU, `jonfelix.apaolaza@ehu.eus`. Lead author of the pilot studies on reversal-aware fine-grained NLI and causal coherence; experience in task design, evaluation, and LLM reasoning analysis.

**Aitor Soroa** (Advisory-Organizer), HiTZ Basque Center for Language Technology – Ixa NLP Group, University of the Basque Country UPV/EHU, `a.soroa@ehu.eus`. Expertise in semantic processing, evaluation, and robust NLP experimentation.

**Rodrigo Agerri** (Advisory-Organizer), HiTZ Basque Center for Language Technology – Ixa NLP Group, University of the Basque Country UPV/EHU, `rodrigo.agerri@ehu.eus`. Expertise in semantic NLP, information extraction, multilingual systems, and modern LLM-based approaches. He will support the multilingual extension and system-oriented analysis.

The organizing team will maintain the task website, mailing list, GitHub repository, Zenodo release, and CodaBench competition; prepare baseline systems and scorers; and manage participant papers and reviewing.

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